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A Computational Syntactic Analysis: Testing Style Hypotheses On The Confessio Patricii Through Dependency Syntax

A Case Study on C. Mohrmann[1] and D. Howlett's[2] Theories of Patrick's Latin

Abstract

This project applies computational, dependency-based syntactic analysis to Confessio Patricii, the fifth-century autobiographical text attributed to Saint Patrick. The goal is to bring quantitative evidence to a long-standing scholarly disagreement about the nature of Patrick's Latin style. The project uses a CoNLL-U annotation of the text. The annotation was produced with UDPipe and manually corrected in Arborator Grew. Two opposing characterizations of Patrick's prose are turned into measurable syntactic indicators. The first is Christine Mohrmann's claim that the prose is simple and paratactic. The second is David Howlett's claim that it conceals a deliberate rhetorical architecture of parallelism and chiasmus. Both indicators follow methods presented in the Computational Philology course. The chiasmus indicator is also adapted from recent computational work on chiasmus detection. The analysis finds partial, direction-consistent support for Mohrmann's hypothesis. It finds weak and inconsistent support for Howlett's hypothesis. The project also shows the methodological limits of testing literary-critical claims through dependency-tree statistics alone.

Introduction

The idea for this project began with a 2025 paper by McGovern, Sirin, and Lippincott, Computational Discovery of Chiasmus in Ancient Religious Text[3]. The paper showed that

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[1] Mohrmann, Christine. The Latin of Saint Patrick: Four Lectures. Dublin: Dublin Institute for Advanced Studies, 1961. https://shop.dias.ie/product/the-latin-of-saint-patrick-four-lectures.

[2] Howlett, David. The Celtic Latin Tradition of Biblical Style. Dublin: Four Courts Press, 1995. https://archive.org/details/celticlatintradi00howl.

[3] McGovern, Andrew, Alexandra Sirin, and Tom Lippincott. "Computational Discovery of Chiasmus in Ancient Religious Text." In Proceedings of the 2025 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL 2025), 2025. https://aclanthology.org/2025.naacl-short.13.pdf.

a structural literary device long studied only through close reading, biblical chiasmus, can be approached computationally instead, by measuring symmetry patterns directly in annotated text. This suggested a broader possibility. Some disputes in philology have historically been settled by literary intuition, and these disputes could perhaps be reframed as testable, reproducible questions once a text is represented as a syntactically annotated corpus. Confessio Patricii offered an unusually well-suited case to try this because the literature on its style already contains a sharply defined, two-sided scholarly debate.

The debate begins with Christine Mohrmann's book, The Latin of Saint Patrick: Four Lectures. The book builds a sustained linguistic argument, moving from the general structure of Patrick's language to its syntax and vocabulary, and then connecting this syntax to the language of the Bible. Mohrmann wrote against an older tradition that treated Patrick as an unskilled writer, one whose Latin showed no literary craft at all. She disagreed with this view, arguing that Patrick's supposed crudeness was actually a coherent variety of Latin, which she called “rustica latinitas”[4]. This is a late, colloquial register, built on simple clause-chaining rather than the periodic, subordination-heavy syntax of classical prose.

The book's first reviewers did not consider the matter closed. Jean Gagnepain[5] reviewed it in Études Celtiques in 1963, and he credited Mohrmann's intent to bring an impartial linguistic contribution to a contested question. But he also doubted that her demonstration was fully convincing. This means the "simple language" thesis stayed a hypothesis rather than an established fact, even at the time of its first publication.

David Howlett proposed the opposite reading of the same texts in The Celtic Latin Tradition of Biblical Style. He rejected the idea of colloquial simplicity and argued that Patrick's prose is carefully organized, placing Patrick within a wider group of early Celtic Latin authors who had deep, continuous engagement with the Vulgate[6]. Howlett claimed their prose uses deliberate biblical rhetorical figures, mainly parallelism and chiasmus, which work at the level of meaning and through the repetition of specific words rather than through surface grammar alone.

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4 "Rustica latinitas" (village/simple Latin) refers to the vernacular or colloquial Latin (sermo rusticus). Unlike the strict literary Latin (latinitas), it was distinguished by a simplified grammar and a large number of dialectal expressions, becoming the basis for the development of the Romance languages.

5 Gagnepain, Jean. Review of The Latin of Saint Patrick: Four Lectures, by Christine Mohrmann. Études Celtiques 10, no. 2 (1963): 595–597. https://www.persee.fr/doc/ecelt_0373-1928_1963_num_10_2_1379_t1_0595_0000_2.

6 The Vulgate is the standard Latin translation of the Bible, created in the late 4th century. The word comes from the Latin vulgata, which literally means "common" or "accepted version." https://www.britannica.com/topic/Vulgate

This reading also met with scholarly skepticism. Alan B. E. Hood[7] reviewed Howlett's book in Early Medieval Europe in 1999, and Gernot Wieland[8] reviewed it in the Journal of Ecclesiastical History in 1997. Both reviews raised methodological doubts about how reliably such patterns could be told apart from coincidence, a concern Christopher Snyder's review raised as well.

This disagreement has not been resolved, and it persists across the original studies and their critical reviews. This persistence motivates the present project and does not try to settle the dispute between Mohrmann and Howlett on literary grounds. Instead, it treats their claims as a source of testable hypotheses, and it asks a simple question. Can dependency-based syntactic analysis of Confessio offer reproducible, quantitative evidence about either position? The analysis follows the methods introduced in this course, and this question leads directly to the research question and the two hypotheses developed below.

Research Question

Can quantitative dependency tree syntactic analysis, applied to Confessio Patricii, provide a testable and replicable basis for the long-standing academic debate about the style of Patrick's Latin? This debate is built around two opposing positions, Mohrmann's thesis of a simple, colloquial, paratactic language, and Howlett's thesis of hidden rhetorical complexity, including parallelism and chiasmus. If such a basis can be provided, the further question is to what extent computational methods confirm, complicate, or refute each of these qualitative interpretations.

H1:The first hypothesis follows directly from Mohrmann's thesis, and it asks to what extent the syntactic structure of Confessio Patricii supports her view once that structure is measured directly. The structure is measured through the ratio of parataxis to hypotaxis, based on the UD labels conj and cc for coordination, against acl, advcl, ccomp, and csubj for subordination. If Mohrmann is right that Patrick's Latin is a simple, additive, colloquial register (rustica latinitas) rather than classical periodic prose with deep subordination, then this ratio should shift systematically toward coordination across the text, so that coordinating relations clearly outweigh subordinating ones in most or all sentences.

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[7] Hood, Alan B. E. "Lighten Our Darkness: Biblical Style in Early Medieval Britain and Ireland." Early Medieval Europe 8, no. 2 (1999): 283–296. https://doi.org/10.1111/1468-0254.00048.

[8] Wieland, Gernot. Review of The Celtic Latin Tradition of Biblical Style, by David Howlett. Journal of Ecclesiastical History 48, no. 4 (1997). https://www.cambridge.org/core/journals/journal-of-ecclesiastical-history/article/abs/celtic-latin-tradition-of-biblical-style-by-david-howlett-pp-xix-400-blackrock-co-dublin-four-courts-press1996-35-1-85182-1430/DE22DDD48415187AFDBF8F3DC9811A0B.

H2: The second hypothesis follows from Howlett's thesis instead, and it takes a more cautious form, since his claim concerns meaning and lexical repetition rather than grammar alone. The question is whether statistically significant traces of hidden parallelism or chiasmus, as Howlett describes them in Patrick's prose, can be detected through the symmetry of part-of-speech sequences (UPOS) within the sentences of Confessio. If such a hidden biblical-style rhetorical structure is present, as Howlett argues, then the symmetry of the UPOS sequence around the midpoint of a sentence should significantly exceed the level of symmetry expected from a random arrangement of the same grammatical categories, and this effect should appear consistently across the text rather than only in a few short sentences, where a chance match is more likely to occur on its own.

Operational Process Of The Project

The project followed a fixed pipeline from raw text to the final quantitative results. The starting point was the Latin text of Confessio Patricii[9], which was first segmented into sentences and tokens. This segmentation was carried out using UDPipe with the latin-ittb-ud model, and the resulting output was then manually corrected in Arborator Grew to fix tokenization, tagging, and dependency errors before any further analysis. The corrected file was loaded as a CoNLL-U structure and flattened into a single per-token table, where every annotation layer, lemma, part of speech, morphological features, and syntactic relation, was stored side by side for each token. A stable identifier was assigned to every token, both a local sentence-based ID and a running global ID across the whole document, so that any result could later be traced back to its exact position in the text. Only after this preparation stage were the syntactic metrics computed, and the two main hypotheses, Mohrmann's parataxis/hypotaxis ratio and Howlett's symmetry-based chiasmus proxy, were built directly on top of this annotated, ID-stable dataset rather than on the raw text.

Before testing either hypothesis, it was useful to look at the corpus as a whole through standard linguistic categories, since this gives a baseline picture of the text against which the more targeted hypotheses can be read. The excerpt analyzed contains 424 tokens across 12 sentences. Nouns are the most frequent part of speech (20.5 percent of all tokens), followed by verbs (13.7 percent) and coordinating conjunctions (10.9 percent), a distribution that already hints at the heavy use of et-coordination later confirmed by the syntactic metrics. At the level of dependency relations, coordinating relations (cc and conj) together account for 84 occurrences, more than any other relation type apart from punctuation, while only 12 tokens carry the root relation, one per sentence as expected. This general profile of the corpus, dominated by nominal and coordinating structures

9 https://www.confessio.ie/etexts/confessio_italian#

rather than by subordinating ones, sets up the more specific question explored next: whether this coordination-heavy profile reflects the kind of simple, paratactic style Mohrmann describes, or whether it coexists with the deeper, more deliberate structure Howlett claims to find beneath the surface.

Hypothesis 1

Introduction. In The Latin of Saint Patrick: Four Lectures, Christine Mohrmann argued against a common view in historiography that treated Patrick as a weak writer whose text lacked any literary devices. Building on the earlier work of Ludwig Bieler, she proposed to test this claim not through literary impression but through a strict linguistic analysis of Patrick's two texts, the Epistola and the Confessio. Her main thesis is that Patrick's Latin belongs to a late colloquial register, rustica latinitas, and that it is built mainly on parataxis, the simple stringing together of clauses through et, rather than on hypotaxis, the complex subordination typical of classical prose such as that of Cicero or Caesar. A modern reviewer of the book, Jean Gagnepain, acknowledged that Mohrmann's goal was to bring "an impartial contribution from a linguist" to a contested question, but he himself expressed doubt that her demonstration was fully convincing. In other words, the thesis about the "simplicity" of Patrick's language was already disputed at the time of publication, rather than an established fact.

Method of measurement. The hypothesis was tested through the parataxis/hypotaxis ratio, that is, the relationship between parataxis and hypotaxis, measured through CoNLL-U syntactic labels. In this work, the method was applied in an expanded form. Subordination was counted using four labels, acl, advcl, ccomp, and csubj, while coordination was counted using two labels, conj and cc. Each of the two measures was calculated as a share of the sentence length, that is, how many words out of a hundred carry that syntactic role. If Mohrmann is right, the share of coordination should be clearly higher than the share of subordination across the whole text.

Verification result. Across the twelve sentences of the text, the mean coordination share was 0.148, and the mean subordination share was 0.049. Coordination turned out to be roughly three times higher than subordination, and this advantage holds in every sentence that contains any clausal relations at all, including the longest sentence, sentence 9 (132 words, 31 coordinating relations versus 11 subordinating ones), and sentence 5 (90 words, 23 versus 4).

How to read these numbers. The results table has several columns, and each one shows a different part of the picture. The column n_subordinate_rel is simply the count of subordinating relations in a sentence, and the column n_coordinate_rel is the count of coordinating relations. These two columns give raw numbers, but raw numbers are hard to compare across sentences of different lengths, so two more columns sit next to them. The column subordination_ratio is the share of subordination, that is, the count of subordinating relations divided by the sentence length. The column coordination_ratio is the same calculation for coordination.

These shares are easier to read as percentages. A share of 0.148 means that, on average, 15 words out of every 100 in a sentence are either the conjunction et or a coordinated element attached to it. A share of 0.049 means that only about 5 words out of 100 belong to a subordinating construction. Comparing the two numbers, for every subordinated word in the text there are, on average, three coordinated words.

Looking at specific sentences makes the picture clearer. In sentence 9, coordination takes up almost a quarter of the sentence length, 31 words out of 132, about 23 percent, which is higher than the document average of 14.8 percent. This means that the longest sentence in the text is also the one most saturated with coordination. The highest coordination value in the whole document belongs to sentence 11, only 13 words long, where coordination reaches 0.308, almost a third of the sentence, even though the sentence itself is very short.

Interpretation and critical analysis. At first glance, the result supports Mohrmann. The parataxis/hypotaxis ratio across the whole text of Confessio is skewed toward parataxis, which fits the picture of a simple, additive style. But two points soften this conclusion.

The first point concerns what the metric itself counts. It only measures the number of subordinating relations, not their depth. Sentence 9 has 11 subordinating relations and, at the same time, the deepest dependency tree in the entire corpus. So even though subordination is numerically outweighed by coordination here, it is structurally quite significant in this particular sentence.

The second point concerns what we are comparing the result against. The ratio of 0.148 was calculated only within the text of Confessio itself, without any comparison to classical Latin. Without a comparison text, it is not possible to say whether this lean toward parataxis is unusually large, or whether subordinating constructions are simply rarer in Latin prose in general, including in Cicero and Caesar.

The hypothesis is therefore only partially confirmed. The direction predicted by Mohrmann, the dominance of parataxis over hypotaxis, is statistically confirmed through the parataxis/hypotaxis ratio. But the thesis of "simplicity" is complicated by the subordination that is still present in the text, especially in its most theologically dense sentence.

Hypothesis 2

Introduction. In The Celtic Latin Tradition of Biblical Style, David R. Howlett proposed a view of Patrick's texts that is opposite to Mohrmann's. Where Mohrmann saw simple, colloquial syntax with no literary devices, Howlett found a hidden rhetorical architecture of a biblical type, inherited through constant reading of the Vulgate. The key devices in his argument are parallelism, the repetition of the same structure in the pattern A-B / A-B, and chiasmus, the mirrored repetition of a structure in the pattern A-B / B-A, named after the shape of the Greek letter χ. According to Howlett, such figures organize composition not only in poetry but also in the Latin prose of early medieval Ireland, including the Confessio. This hypothesis is methodologically riskier than H1, because parallelism and chiasmus, as Howlett describes them, work at the level of meaning and lexical choice, that is, the repetition of specific words and ideas, rather than at the level of grammar, and they are therefore difficult to operationalize directly through CoNLL-U without significantly simplifying the original idea.

Method of measurement. The method was built in analogy with a recent 2025 paper that addresses exactly this task on biblical texts[10]. The idea is to replace Howlett's semantic chiasmus with a rough syntactic analogue. Each sentence is split into a first half and a second half by part of speech, that is, by UPOS tags such as VERB, NOUN, and ADJ. Two types of matches are then checked. The first checks whether the first half matches the second half in the same order, and this is the test for parallelism. The second checks whether the first half matches the second half in reverse, mirrored order, and this is the test for chiasmus. Whichever version produces the higher match is taken as the sentence's mirror_score.

What each column in the table means. The column sent_id is the sentence number in the text. The column n_tokens is the length of the sentence in words. The column mirror_score is the main number in this table. It is the share of matching positions between the two halves of the sentence, ranging from 0, no matches at all, to 1, a complete match. The higher the number, the more symmetrical the sentence's grammatical structure. The column pattern_type shows which type of symmetry turned out to be stronger. A value of parallelism-like means the direct order wins, and a value of chiasmus-like means the mirrored order wins.

What the table showed. The highest mirror_score values came from sentence 2 (0.400) and sentence 3 (0.333), but both of these sentences are very short, only 10 and 6 words. In the longest sentence of the text, sentence 9, which is devoted to the Trinity and contains

[10] McGovern, Sirin, Lippincott, Computational Discovery of Chiasmus in Ancient Religious Text, NAACL 2025

132 words, the score is moderate, 0.182. The lowest result, zero, belongs to sentence 10, which is also very short, 6 words.

Interpretation and critical analysis. At first glance, the result could be read as weak support for Howlett's idea, since some symmetry in the grammatical structure of the sentences does exist, especially in the short sentences 2 and 3. But two important reservations apply here.

The first reservation concerns the reliability of the numbers themselves. The highest values came precisely from the shortest sentences, 2, 3, and 11. This is to be expected and is not very informative, because the shorter a sentence is, the fewer ways its words can be arranged, and the easier it is to get a high match purely by chance, without any deliberate intent on the author's part. In the longest and most substantively important sentence of the text, sentence 9, the symmetry was only moderate, 0.182, and this is a far more reliable signal, because at that length a chance match on its own is highly unlikely.

The second reservation concerns the fact that the method does not test quite what Howlett actually claims. His argument rests on the repetition of specific words and ideas, for example the phrase custodiuit me... muniuit me... consolatus est me from sentence 5, rather than on an abstract symmetry of parts of speech. The mirror_score method is a simplified proxy. It can miss a genuine lexical chiasmus in Howlett's sense, and at the same time it can mistakenly count a coincidental match of grammatical categories as a "chiasmus-like" pattern, even when the words involved have no real connection in meaning.

Unlike H1, where the direction of the effect, a clear dominance of coordination over subordination, was stable across the whole text and easy to interpret, the result for H2 is uneven and weakly expressed. Symmetry of parts of speech is present in Confessio, but it is small, concentrated mainly in short sentences, and does not provide confident support for Howlett's hypothesis in its original, meaning-based sense. A more honest test of the original idea would require an analysis of the repetition of specific lemmas and concepts, rather than just the sequence of parts of speech.

Conclusion

This project set out to test whether dependency-based syntactic analysis could offer reproducible, quantitative evidence relevant to the long-standing disagreement between Mohrmann and Howlett over the style of Patrick's Latin. The analysis was carried out on a partial excerpt of Confessio Patricii, not the complete text, so the conclusions below describe a sample of the work rather than the whole composition.

For Mohrmann's hypothesis, the parataxis/hypotaxis ratio gave consistent results. Coordination outweighed subordination in every sentence of the excerpt that contained clausal relations, and the average coordination ratio was roughly three times higher than the average subordination ratio. This supports the direction Mohrmann predicted, and it does so steadily across short and long sentences alike. At the same time, the longest sentence in the excerpt also had the deepest dependency tree, so the subordination that does occur is structurally significant even though it is numerically rare. The hypothesis is therefore only partially confirmed. The text leans toward paratactic coordination, but it is not free of complex subordination, especially in its most theologically dense passages.

For Howlett's hypothesis, the picture was much weaker. The symmetry measure built around UPOS sequences showed some signal, but it was uneven across sentences, and the strongest results came from the shortest sentences, where random chance is more likely to produce an apparent pattern. The longest sentence in the excerpt showed only a modest level of symmetry, and the shortest sentence showed none at all. The method also tests a simplified version of Howlett's claim, since his argument rests on the repetition of specific words and ideas, not on the grammatical category of each word. The result is a weak and uneven finding, not a confirmation of Howlett's reading.

Taken together, the two hypotheses received different levels of support from the same computational pipeline. The Mohrmann-style measurement aligned well with the kind of syntactic feature it was designed to capture, while the Howlett-style measurement struggled to capture a phenomenon that is, by its own definition, more lexical and semantic than grammatical. This asymmetry is itself a useful finding. It shows that some literary-critical claims translate more naturally into computational tests than others, and that the success of a computational method depends heavily on how closely its proxy matches the original claim.

This pattern is also visible at the level of the corpus as a whole, not only in the two hypotheses taken separately. The general linguistic profile of the excerpt, dominated by nouns, verbs, and coordinating conjunctions, with coordinating relations far more frequent than any subordinating ones, already pointed toward a coordination-heavy style before either hypothesis was tested in detail. The subsequent results refined this picture rather than overturning it: coordination indeed dominates, as both the corpus profile and the H1 results show, but the text's most complex sentence, sentence 9, still carries the deepest subordination and only a moderate level of grammatical symmetry, suggesting that simplicity and complexity coexist in the same passages rather than describing two separate parts of the text.

Limitations

The analysis was conducted on a partial excerpt of Confessio Patricii, not the full text. The results describe the patterns found in this excerpt, and they should not be read as a complete characterization of the whole work.

The excerpt itself is short, twelve sentences in total, so the statistics are based on a small sample. A few unusual sentences can shift the averages noticeably, and some metrics, especially the symmetry score for Howlett's hypothesis, are sensitive to sentence length in ways that make short sentences less reliable than long ones.

Neither hypothesis was tested against a comparison text written in classical Latin. Both metrics show how Confessio behaves on its own terms, but they do not show whether its values are unusually high or low compared to authors like Caesar or Cicero. Without this comparison, it is hard to say whether the patterns found are specific to Patrick or typical of Latin prose more broadly.

The symmetry measure used for Howlett's hypothesis works only at the level of part-of-speech tags. It does not look at lemmas, word meaning, or thematic repetition, which are the actual basis of Howlett's argument. This means the measure can miss real instances of lexical chiasmus, and it can also register a coincidental match between grammatical categories as a false positive.

The annotation itself depends on a single CoNLL-U file, produced by UDPipe and corrected by hand in Arborator Grew. Manual correction reduces errors, but it does not remove the possibility of remaining mistakes or debatable analytical choices, particularly in long and syntactically complex sentences such as sentence nine.

Finally, both metrics are descriptive rather than causal. They show patterns in the syntactic structure of the text, but they cannot determine whether any pattern found was intentional on Patrick's part, accidental, or a natural feature of the Latin he had available to him.

Perspectives For Future Research

Several extensions should be deepened, namely what would more decisively separate Mohrmann's and Howlett's diametrically opposed positions. A classical-Latin comparator text, processed through alignment algorithms, would turn the current internal asymmetry into a relative claim against known classical style, which matters for both positions at once: it would show whether Patrick's coordination is unusually high (supporting Mohrmann) or whether his apparent symmetry is unusually low or high relative to other Latin authors (bearing on Howlett). Computational stemmatology could test the same metrics across the different manuscript witnesses of Confessio, showing whether the coordination-heavy profile is stable or shifts through scribal transmission.

Two further directions were explored only briefly in earlier, exploratory versions of this project and were ultimately set aside, but they remain relevant here precisely because they target the weakest point of the current H2 test, the gap between syntactic proxies and Howlett's actual lexical and semantic claim. A lemma-repetition measure, following the stylometric distinction between feature, marker, and styleme, would track repeated words and concepts at symmetric positions in a sentence, such as the me-repetition noted in sentence 5, rather than repeated part-of-speech categories, bringing the chiasmus test much closer to what Howlett actually argues. A second, more exploratory direction tested whether Patrick's vocabulary showed unusual letter-based numerical coincidences , an idea raised in Howlett's broader argument. This measure produced no signal beyond what a random-letter baseline would predict, but a more refined version, testing alternative letter-to-number systems or restricting the search to theologically salient terms, could clarify whether this lack of signal reflects the absence of the phenomenon or simply the limits of the specific encoding tested.

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